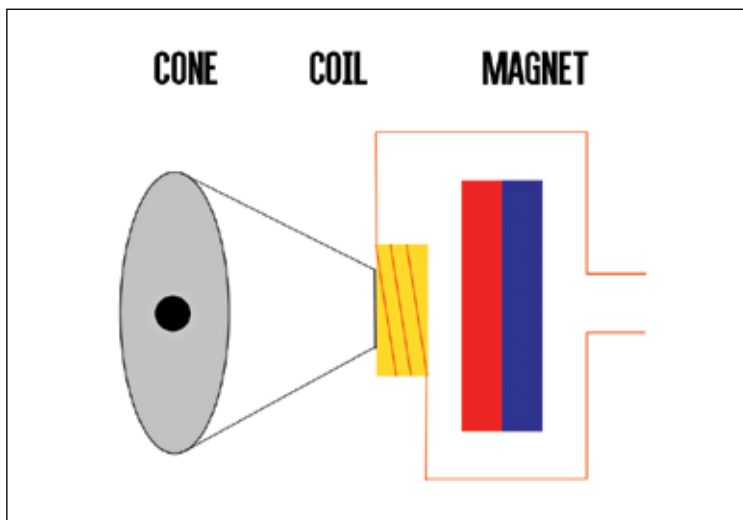




A simple electrodynamic loudspeaker layout

the electromagnet, the magnetic field's direction is changed rapidly. Due to this rapid change, it is rapidly attracted to and repelled from the permanent magnet. There are two characteristics of the vibrations that in turn affect two characteristics of the sound produced - the frequency determines the pitch of the sound produced, and the amplitude governs the volume – if you turn the volume up high enough, you might even be able to see the diaphragm covering the cone move.

Again, logically, the material of the electromagnet must respond well to quick changes in vibration, and be able to respond to the entire range of audible frequencies. It must also be sturdy and pliable. Hence, the loudspeaker electromagnet is usually made of copper, though aluminium may also be used. The advantage of aluminium is its light weight, which allows it to respond more easily to higher frequencies. Disadvantages include its toughness in being soldered, hence requiring connections to be crimped and sealed.

The permanent magnet, on the other hand, has only one simple requirement: its magnetic field must be strong enough to produce a high strength magnetic field without any external charging or support. Earlier designs used electrically powered field coils. But with the emergence of stronger easy to use permanent magnets, Alnico, an alloy, became the most ubiqui-